On the (co-) homology of commutative rings

Daniel Quillen[1]

This paper is devoted to a summary of the main results of a (co-) homology theory of commutative rings due independently to André [1] and the author [19]. The theory has reached a degree of completeness that there is little doubt that it is the correct cohomology theory for commutative rings. It is perhaps worthwhile to relate this theory to definitions made by other authors.

For any morphism $A \to B$ of commutative rings and B-module M, let $\mathrm{Der}_A(B, M)$ be the B-module of A-algebra derivations of B with values in M, and let $\mathrm{Exalcomm}_A(B, M)$ be the B-module of infinitesimal A-algebra extensions of B by M ([11, Chapter IV]). It is known that the functors Der and Exalcomm possess the following two properties:

0.1. (*Transitivity*). Given morphisms $A \to B \to C$ of commutative rings and a C-module M there is a six-term exact sequence

$$0 \to \mathrm{Der}_B(C, M) \to \mathrm{Der}_A(C, M) \to \mathrm{Der}_A(B, M)$$
$$\to \mathrm{Exalcomm}_B(C, M) \to \mathrm{Exalcomm}_A(C, M) \to \mathrm{Exalcomm}_A(B, M).$$

0.2. (*Flat base change*). Given morphisms $A \to B$, $A \to A'$ of commutative rings and an $A' \otimes_A B$ module M, there are isomorphisms

$$\mathrm{Der}_{A'}(A' \otimes_A B, M) \simeq \mathrm{Der}_A(B, M)$$

$$\mathrm{Exalcomm}_{A'}(A' \otimes_A B, M) \simeq \mathrm{Exalcomm}_A(B, M) \quad \text{if } \mathrm{Tor}_1^A(A', B) = 0.$$

The cohomology theory to be presented here associates to each morphism $A \to B$ of commutative rings and B-module M cohomology B-modules $D^q(B/A, M)$ for each integer $q \geq 0$, coinciding with $\mathrm{Der}_A(B, M)$ and $\mathrm{Exalcom}_A(B, M)$ for $q = 0$ and 1 respectively, and extending in the obvious way the properties 0.1–0.2.

In [13] Harrison gave a definition of commutative ring cohomology using a

[1] Supported by NSF GP-6959 and the Alfred P. Sloan Foundation.

subcomplex of the Hochschild complex for computing associative algebra cohomology. Harrison restricted himself to the case where the ground ring A is a field; his definition does not extend to a general morphism $A \to B$ since one needs at least that B be projective as an A-module in order that Harrison's H^2 coincide with $\mathrm{Exalcomm}_A(B, M)$. However, it can be shown (§9 below) that Harrison's complex may be used to compute the correct cohomology $D^*(B/A, M)$ when B is projective as an A-module and when B is of characteristic 0, that is, an algebra over the rational numbers.

In [**16**], Lichtenbaum and Schlessinger define a satisfactory cohomology theory in dimensions $q \leq 2$ in the sense that their cohomology extends the exact sequence of 0.1 to nine terms. Their method uses a free differential graded anticommutative A-algebra resolution of B. It may be shown that their cohomology group $T^q(B/A, M)$ coincides with our $D^q(B/A, M)$ for $q \leq 2$ and that free differential graded anticommutative A-algebra resolutions of B may be used to compute $D^*(B/A, M)$ when B is of characteristic zero (§9).

In [**21**] it is shown how the transitivity and flat base change properties of the Lichtenbaum-Schlessinger $T^q(B/A, M)$ for $q \leq 2$ (hence also of our $D^q(B/A, M)$) yield a satisfactory deformation theory for commutative algebras. Moreover the Lichtenbaum-Schlessinger cohomology is known to be the same as the theory of Gerstenhaber [**8**] specialized to commutative rings.

The cohomology groups $D^*(B/A, M)$ are defined as suitable nonabelian derived functors of the functor $X \mapsto \mathrm{Der}_A(X, M)$ on the category of A-algebras over B, where the derived functors are defined by applying the functor dimensionwise to a free (semi-) simplicial A-algebra resolution of B and taking the cohomology of the associated cosimplicial B-module. A particular example of such a resolution is furnished by the standard construction associated to the (sets)—(A-algebras) cotriple. Thus the cohomology $D^*(B/A, M)$ is a kind of cotriple cohomology [**5**], however, from the cotriple point of view one loses sight of the possibility of choosing a resolution for computing the derived functors with special properties, a technique vital for proving the transitivity and flat base change properties of the commutative ring cohomology.

1. **Derived functors.** In a category closed under finite limits and having enough projective objects, it is possible to define left derived functors of any functor from the category to an abelian category generalizing those of Dold-Puppe [**7**]. We give the construction of these derived functors in this section.

An object P of a category $\mathscr{C}$ is said to be *projective* if for any effective epimorphism $p: X \to Y$, $\mathrm{Hom}(P, p): \mathrm{Hom}(P, X) \to \mathrm{Hom}(P, Y)$ is surjective. $\mathscr{C}$ is said to have *enough projective objects* if for every object X there is an effective epimorphism $P \to X$ with P projective. When $\mathscr{C}$ is closed under finite projective limits and has enough projective objects, then the effective epimorphisms are the maps p for which $\mathrm{Hom}(P, p)$ is surjective for all projective objects P [**18,** II, §4, Proposition 2]; in particular the effective epimorphisms are universally effective.

Let $s\mathscr{C}$ be the category of (semi-) simplicial objects over $\mathscr{C}$. We always identify an object X of $\mathscr{C}$ with the constant simplicial object whose simplicial operators are